

Culturomics Phenomenon

Through the Wikipedia article on Michael Jackson

A 2009-2026 case study, building on J.B. Michel (2011)

INTRODUCTION

The Culturomics Phenomenon

J.B. Michel (2011) showed that a society's collective attention to events leaves measurable traces in digital sources, and follows quantifiable patterns.

“...an initial response to an exogenous shock, followed by a long-term decay reflecting the process of collective forgetting”

— Michel J.B., 2011

Michel's original source

Google Books corpus
40+ million books, n-gram frequency

Critique (Pechenick J.P. , 2015)

The corpus is library-like: one book equals one vote, regardless of actual readership. A growing share of scientific texts distorts word frequencies.

Why Wikipedia

01

Michel already used Wikipedia

Fame analysis of 740,000 people in the original 2011 paper

02

Two independent signals

Views measure consumption, edits measure production. Books offer only one signal

03

Real time

Edits are recorded within hours. Books require months to be published

04

Collectiveness

Decentralized editing is a more direct embodiment of Halbwachs's collective memory

Google Books

text publication

one signal

years of delay

Wikipedia

consumption + production

two signals

real time

THEORETICAL FRAMEWORK

Three Phases Of The Phenomenon

2001 – JUNE 2009

Before the shock

A stable plateau. Slow article growth, routine maintenance mode.

25 JUNE 2009

The flashpoint

A phase transition. Exogenous shock: a surge in views and edits.

AFTER 2009

The residue

Views and edits decay. Article size never returns to baseline.

Data And Method

Pageviews

pagecounts-raw (2009)
+ Pageviews REST API (2015-2026)

4,091 daily data points

Edits

MediaWiki Action API
prop=revisions, 2001-2026

33,893 revisions

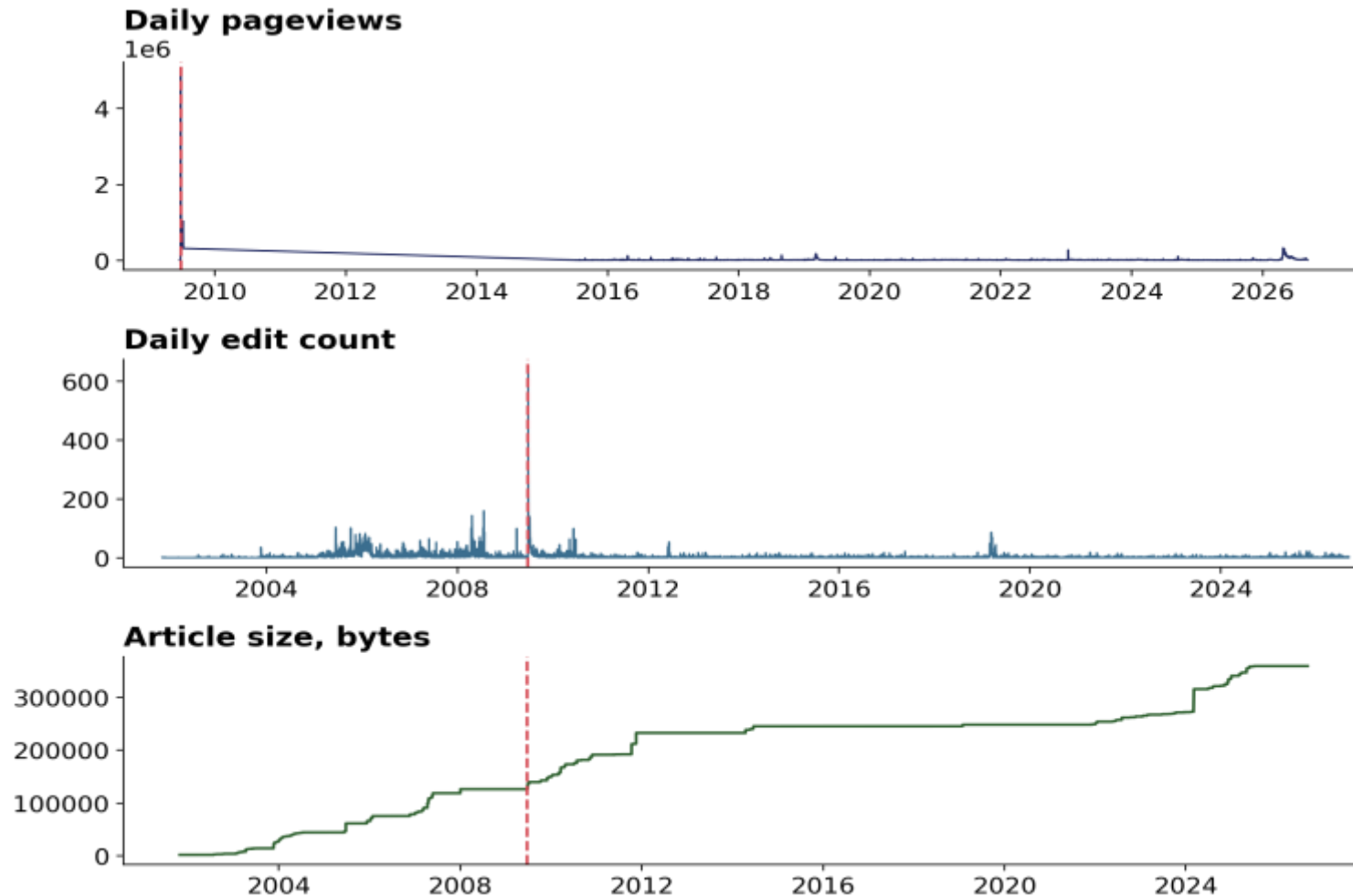
Method

Michel's half-life:
peak - first day below half

No curve fitting

LAYER 1 DESCRIPTIVE

The Overall Picture



All three signals mark the same moment, 25 June 2009, but behave very differently afterwards.

Pageviews

12,747 to 4,972,223 in one day

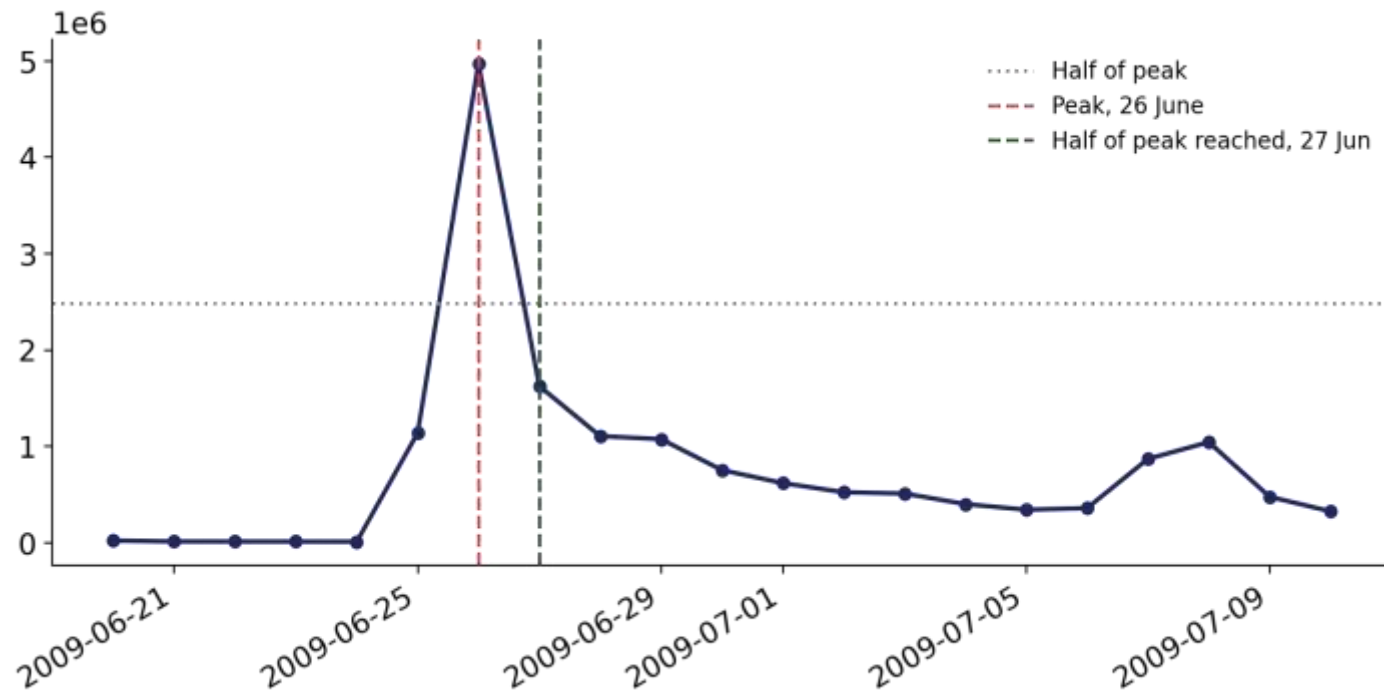
Edits

5 to 642 per day

Article size

never returns to baseline

How Fast Attention Faded



1

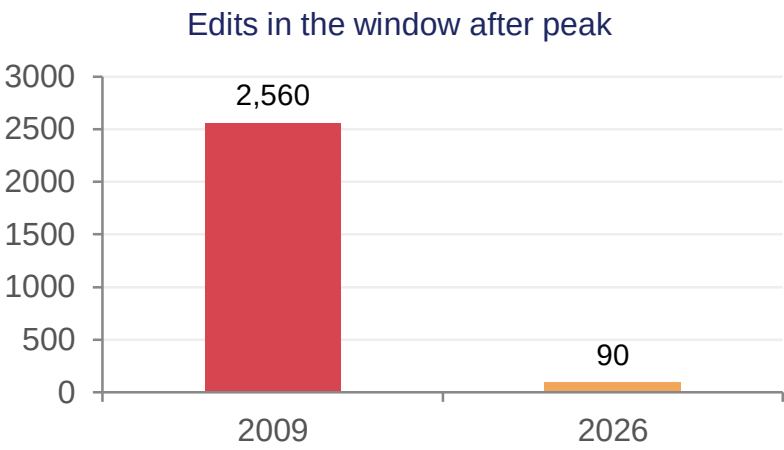
day to reach half of peak

The acute shock phase ends within a day, but a full return to baseline takes about two weeks.

2009 Versus 2026

The artist's death and the release of a movie: two attention spikes of a different kind

	2009 (death)	2026 (film)
Peak pageviews	4,972,223	339,354
Edits in window	2,560	90
Change in size	+13,347 bytes	0 bytes



The difference reflects not only timing but the nature of the event: a death demands a rewrite of the facts, a film release only their addition.

LIMITATIONS

Limitations

- cummax understates size change near the article's historical maximum
- The 2009 and 2026 comparison rests on only two cases, not a statistical sample
- Some hourly files in the 2009 archive were unavailable on the server

Conclusion

Views and edits fall. Article size does not.

Culturomic analysis, originally built on Michel's book corpus, applies equally well to Wikipedia data, with the added benefit of separating consumption and production signals. Collective memory crystallizes not at the moment of attention, but at the moment of editing.

